

PERSPECTIVE

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Twenty-five years of septic shock hemodynamic resuscitation trials: a conceptual perspective

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Abstract

Background Twenty-five years of septic shock resuscitation trials have progressively transformed the physiological understanding of shock and the way bedside interventions are conceptualized. Early strategies focused on correcting isolated global hemodynamic or metabolic targets through protocolized interventions, whereas more recent approaches increasingly recognize septic shock as a heterogeneous, dynamic, and context-dependent condition requiring individualized physiological assessment.

Main body This perspective discusses how hemodynamic resuscitation trials during this quarter of a century have reshaped both the physiological understanding and methodological foundations of early shock management. Initial strategies, exemplified by early goal-directed therapy, aimed to correct global oxygen-derived variables through protocolized interventions designed to optimize oxygen delivery. However, the lack of reproducibility of this approach in subsequent multicenter trials, together with the recognition of the limitations of several other variables as potential resuscitation targets, revealed that septic shock cannot be adequately addressed through rigid algorithms based on isolated endpoints. Subsequent progress incorporated peripheral perfusion assessment, systematic evaluation of fluid responsiveness, critical care echocardiography, and hemodynamic phenotyping. Within this trajectory, ANDROMEDA-SHOCK shifted attention toward capillary refill time as a rapidly responsive clinical perfusion signal, contributing to improve some outcomes, and potentially limiting over-resuscitation. ANDROMEDA-SHOCK-2 further operationalized a personalized strategy based on capillary refill time, sequential phenotyping, reversible hemodynamic tests, and serial reassessment.

Conclusions A quarter of a century of septic shock resuscitation trials reveal a consistent pattern: strategies centered on fixed and isolated physiological targets have repeatedly failed to achieve reproducible improvements in patient-centered outcomes. Recent data support a shift from isolated hemodynamic or metabolic targets, toward a physiology-guided, phenotype-driven resuscitation strategy, in which interventions are applied as reversible tests

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within short decision cycles and guided by rapidly responsive perfusion signals. Trials such as ANDROMEDA-SHOCK-2 represent important steps in this ongoing transition, but further refinement, validation, and implementation across diverse settings remain necessary.

Keywords Septic shock, Resuscitation, Perfusion, Capillary refill time, Lactate, Fluid responsiveness, Personalization, Phenotypes

Introduction

Twenty-five years of septic shock resuscitation trials have progressively refined the physiological reasoning that guides hemodynamic resuscitation and the way bedside interventions are conceptualized. Early strategies focused on correcting isolated global hemodynamic or metabolic targets through protocolized interventions, whereas more recent approaches increasingly recognize septic shock as a heterogeneous, dynamic, and context-dependent condition requiring individualized physiological assessment. During this period, pivotal randomized controlled trials (RCTs) evaluated diverse resuscitation approaches [1–9]. Rather than representing isolated successes or failures, these trials collectively contributed to a progressive refinement of resuscitation concepts, bedside monitoring tools, and clinical decision-making frameworks.

In this review, we provide a conceptual perspective on the evolution of septic shock hemodynamic resuscitation over the last 25 years, focusing on 6 landmark trials that represent relevant contributions and stepping stones in the development of our current understanding of contemporary septic shock resuscitation (Figs. 1, 2 and Table 1) [1–6].

Historical and physiological evolution of septic shock resuscitation

The rise and fall of ScvO₂ as a resuscitation target

Twenty-five years ago, Rivers et al. published the Early Goal-Directed Therapy (EGDT) trial, which targeted ScvO₂ by manipulating oxygen delivery (DO₂) in the protocolized intervention arm [1]. The study was conducted in a single-center emergency department (ED) and essentially recruited septic patients admitted with protracted septic shock and/or profound hypovolemia. EGDT was associated with a significant survival benefit as compared to usual care, representing the first successful trial specifically targeting a perfusion-related variable. These findings led to the addition of some aspects of the EGDT algorithm into the emerging Surviving Sepsis Campaign (SSC) guidelines [10].

However, after initial enthusiasm, EGDT acceptance declined over the next decade due to several concurrent factors. First, later studies conducted in both ED and intensive care unit (ICU) settings could not reproduce the finding of the very low ScvO₂ values reported

in the EGDT trial [11]. Second, it was realized that ScvO₂ is a non-dichotomous variable, since low and supra-normal values may signal different pathophysiological conditions, but being both associated with increased mortality [12]. Third, the EGDT algorithm relied on debatable physiological principles, such as aiming for an abnormally elevated central venous pressure (CVP), and excessively liberal transfusion thresholds and fluid administration rates [1]. Additionally, growing literature on the risk of fluid accumulation highlighted the need for more balanced fluid resuscitation strategies [13]. Furthermore, three more contemporary multicenter RCTs testing EGDT, although in markedly different contexts, yielded neutral results, thus challenging its external validity [7–9]. This led to an abandonment of ScvO₂ as a resuscitation target in recent SSC guidelines [14], but ScvO₂ monitoring is still recommended as part of a multimodal perfusion monitoring according to the specific resuscitation context [15]. Indeed, the role of ScvO₂ as a valuable physiological signal when interpreting DO₂/VO₂ relationships in shock patients and eventually guiding therapeutic decisions should not be neglected.

The lactate paradox: hypoxia marker but inadequate resuscitation goal

Lactate is a fundamental physiological signal that is traditionally viewed as a marker of tissue hypoxia. Therefore, it has been considered central to septic shock diagnosis and risk stratification, but paradoxically lactate appears to be an unreliable stand-alone therapeutic target, essentially because it has been misread as a direct measure of “oxygen debt” [16–20]. A more precise interpretation is provided by the concept of biphasic lactate kinetics in septic shock [21]. During the early phase, usually within the first 6–12 h, lactate may remain predominantly flow-dependent and therefore responsive to interventions that restore DO₂, perfusion pressure, and microcirculatory flow. In this window, lactate kinetics can provide useful information on the adequacy of initial resuscitation. Thereafter, its determinants become increasingly heterogeneous, including persistent production, impaired clearance, adrenergic-driven aerobic glycolysis, inflammatory metabolic reprogramming, and mitochondrial dysfunction [17–20]. Thus, beyond the early resuscitation phase, persistent hyperlactatemia should not be assumed to represent ongoing global dysoxia, particularly when

Evolution of Septic Shock Resuscitation Concepts: From Single Targets to Personalized Physiology-Guided Therapy

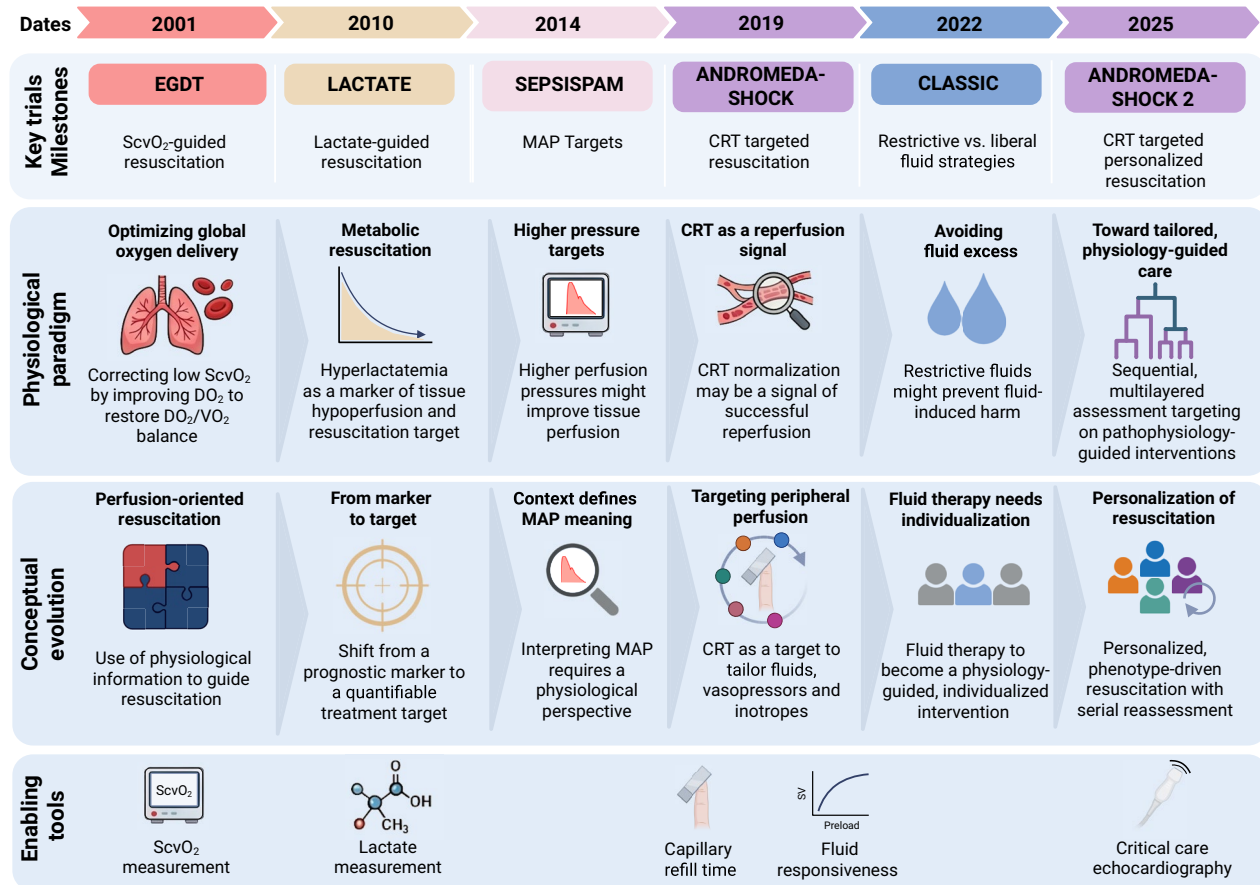


Fig. 1 Evolution of septic shock resuscitation concepts: from single targets to personalized physiology-guided therapy. The figure illustrates landmark trials from the past 25 years, highlighting their key milestones in septic shock resuscitation and the associated evolution of physiological paradigms and conceptual frameworks. Early goal-directed therapy (EGDT) emphasized optimization of global oxygen delivery using central venous oxygen saturation targets. The LACTATE trial incorporated lactate as a marker and therapeutic target for tissue hypoperfusion. SEPSISPAM, evaluating mean arterial pressure targets, highlighted the importance of clinical context when interpreting perfusion pressure goals. ANDROMEDA-SHOCK focused on peripheral perfusion, particularly capillary refill time, to guide fluids, vasopressors, and inotropes, while the restrictive-fluid CLASSIC trial emphasized avoidance of fluid excess and the need to individualize fluid therapy. ANDROMEDA-SHOCK 2 represents a further step toward tailored, physiology-guided care, integrating peripheral perfusion-targeted therapy with sequential, multilayered assessment and serial reassessment of individual pathophysiology. CLASSIC, *Conservative versus Liberal Approach to Fluid Therapy of Septic Shock in Intensive Care*; CRT, capillary refill time; DO₂, oxygen delivery; EGDT, early goal-directed therapy; MAP, mean arterial pressure; ScvO₂, central venous oxygen saturation; SEPSISPAM, Sepsis and Mean Arterial Pressure trial

peripheral perfusion and other hypoperfusion markers have normalized [16, 22, 23].

Nevertheless, the clinical need for a quantifiable early resuscitation endpoint, prompted attempts to convert lactate kinetics into an explicit treatment target. The most developed expression of this approach was the LACTATE trial [2], targeting a $\geq 20\%$ lactate decrease every 2 h during an 8-h intervention period. The trial showed a positive effect on some secondary outcomes, including lower hospital mortality and early improvement in Sequential Organ Failure Assessment (SOFA) score. However, the intervention arm received more fluids and vasodilator support, yet lactate kinetics were essentially unchanged, and the prespecified lactate-reduction target

was achieved equally in both groups, undermining the premise that the protocol meaningfully “treated” lactate [2].

The limited adoption of the LACTATE trial therefore reflects a more fundamental physiological constraint: beyond the early resuscitation window, lactate is a time-dependent and mechanistically heterogeneous signal that is not reliably *modifiable* by further hemodynamic escalation. Additional drawbacks include the complexity of interpreting persistent hyperlactatemia [16, 22, 23], as well as its inadequate kinetics of response to acute hemodynamic interventions, eventually increasing the odds of over-resuscitation.

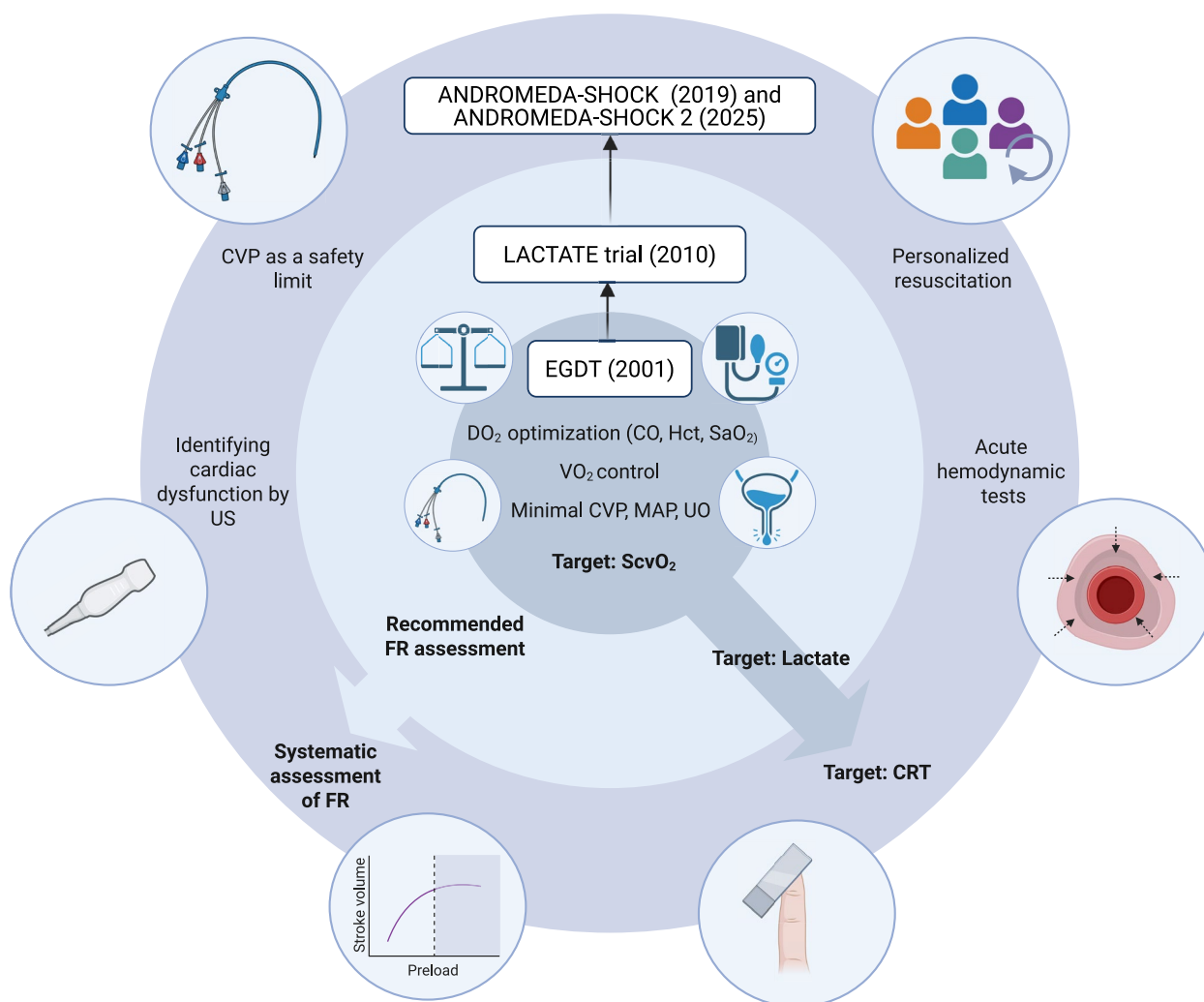


Fig. 2 Conceptual evolution of hemodynamic resuscitation targets in septic shock over the last 25 years. The figure traces the evolution of perfusion- and oxygenation-derived resuscitation targets across four landmark trials. At its core, early goal-directed therapy (EGDT) appears not as a single-criterion strategy but as an already multi-parametric, physiology-oriented approach: it integrated optimization of oxygen delivery (cardiac output, hematocrit, arterial oxygen saturation) and control of oxygen consumption, together with conventional macrohemodynamic boundaries (CVP, MAP, urine output), all directed toward central venous oxygen saturation ($ScvO_2$) as the target. The LACTATE trial then adopted lactate as a treatable hypoperfusion signal (target: lactate), and ANDROMEDA-SHOCK introduced capillary refill time as a rapidly responsive peripheral perfusion target (target: CRT) while incorporating systematic fluid responsiveness assessment. ANDROMEDA-SHOCK-2 expanded this framework by combining CRT-targeted resuscitation with hemodynamic phenotyping, echocardiography, and individualized therapeutic tests, representing an early step toward personalized resuscitation. Surrounding elements depict complementary concepts that matured over the same period, such as the reappraisal of CVP from a target to a safety limit, identification of cardiac dysfunction by ultrasound, and acute hemodynamic testing. Because the figure specifically traces perfusion- and oxygenation-derived target selection, it depicts the four trials that shaped this trajectory. SEPSISPAM and CLASSIC, which addressed mean arterial pressure targets and fluid-volume strategies rather than perfusion-target selection, fall outside this narrower scope and are not shown; both are nonetheless part of the six landmark trials examined in the main text and summarized in Fig. 1 and Table 1. CO, cardiac output; CRT, capillary refill time; CVP, central venous pressure; DO_2 , oxygen delivery; EGDT, early goal-directed therapy; FR, fluid responsiveness; Hct, hematocrit; MAP, mean arterial pressure; SAO_2 , arterial oxygen saturation; $ScvO_2$, central venous oxygen saturation; UO, urine output; US, ultrasound; VO_2 , oxygen consumption

Despite these caveats, lactate remains a relevant monitor to assess the general response of septic shock patients to early hemodynamic resuscitation or source control. A rapid decreasing lactate suggests a favorable response, and conversely, persistent hyperlactatemia independent of the predominant source is associated with worse prognosis and was recently proposed as a criterion to define refractory septic shock [24].

Capillary refill time: more than just “back to basics”

The intrinsic limitations of $ScvO_2$ and lactate as resuscitation goals directed attention to more dynamic perfusion targets such as capillary refill time (CRT). Early observational studies showed the better prognosis associated with rapid CRT normalization [25], and its favorable kinetics of response to hemodynamic resuscitation [26].

Table 1 Physiological rationale, potential drawbacks, and methodological contributions of key hemodynamic resuscitation strategies in septic shock

Strategy (Trial)	Physiological rationale	Potential drawbacks	Methodological contributions
ScvO ₂ -guided resuscitation (EGDT, 2001)	- ScvO₂ reflects balance between DO₂ and VO₂ - Low ScvO₂ signals inadequate DO₂ → correctable by fluids, blood transfusion, or inotropes - Structured, early protocolized approach to perfusion restoration in ED patients - Pioneered concept of targeting perfusion-related variables in septic shock RCTs	- ScvO₂ is non-dichotomous: supranormal values also carry increased mortality - Observational studies failed to reproduce extremely low ScvO₂ in ICU-admitted patients - Algorithm promoted harmful intervention: elevated CVP targets, liberal transfusion, aggressive fluid loading - Three multicenter RCTs (ProCESS, ARISE, ProMISe) failed to replicate survival benefit “One-size-fits-all” approach ignores hemodynamic phenotype heterogeneity	- First RCT to demonstrate survival benefit using a perfusion-related variable as a resuscitation endpoint - Integrated into the Surviving Sepsis Campaign and subsequent resuscitation guidelines - Underscored importance of multicenter replication, single-center RCT results may not be generalizable
Lactate-targeted resuscitation (LACTATE trial, 2010)	- Hyperlactatemia interpreted as surrogate of tissue hypoxia - Strong prognostic signal: lactate > 2 mmol/L associated with higher mortality in septic shock - Targeting ≥ 20% lactate decrease every 2 h provided a measurable, dynamic resuscitation goal	- Lactate determinants are mixed beyond early resuscitation: adrenergic aerobic glycolysis, impaired clearance, stress metabolism - Lactate kinetics is slow and mechanistically heterogeneous → poor real-time feedback for hemodynamic interventions - Prespecified reduction target achieved equally in both trial arms - Intervention arm received more fluids and vasodilators without meaningful lactate improvement → over-resuscitation risk	- Highlighted risk of designing trials around targets not reliably modifiable by the proposed intervention - Stimulated refinement of perfusion endpoint selection criteria for future RCTs
MAP-targeted resuscitation (SEPSIS-PAM, 2014)	- MAP is the principal determinant of organ perfusion pressure; a higher target might better sustain flow to vital organs - Chronic hypertension right-shifts the autoregulation curve, so higher targets (80–85 mmHg) might protect organ perfusion in these patients - Tested whether a universal MAP target of 65–70 mmHg is adequate or should be individualized by premorbid physiology	- MAP is a macrohemodynamic surrogate: identical values can reflect very different microcirculatory and flow states - Achieving a higher MAP requires greater vasopressor exposure, with attendant risks (higher incidence of atrial fibrillation in the high-target arm) - A fixed numeric target ignores individual autoregulatory thresholds and hemodynamic phenotype - Neutral primary outcome (no difference in 28- or 90-day mortality)	- Demonstrated the value of pre-specified subgroup analysis: patients with chronic hypertension required less renal-replacement therapy under the higher target - Quantified the safety trade-off of vasopressor intensity (arrhythmia signal), informing the risk–benefit of pressure escalation - Moved the field from a single universal MAP target toward context- and phenotype-dependent perfusion-pressure goals
CRT-targeted resuscitation (ANDROMEDA-SHOCK, 2019)	- CRT reflects skin microcirculatory perfusion, adrenergic tone, and microvascular reactivity - Rapid kinetics of CRT response to hemodynamic interventions enables real-time resuscitation feedback - Standardized technique reduces inter-rater variability - CRT normalization as a stopping signal may prevent fluid over-resuscitation - Costless, non-invasive, universally available perfusion target	- Physiology behind CRT complex and incompletely understood - CRT non-response may reflect endothelial dysfunction, abnormal microvascular reactivity, or venous congestion - The used algorithm limits individualized hemodynamic phenotyping and echocardiographic integration 28-day mortality difference was non-significant (p = 0.06); benefit driven mainly by organ dysfunction outcomes	- Introduced systematic FR assessment before each fluid bolus as a mandatory safety step in RCT design - Pioneered concept of “acute hemodynamic tests” as reversible, decision-making interventions - Bayesian re-analysis supported CRT superiority, demonstrating value of pre-planned secondary analyses in underpowered trials

Table 1 (continued)

Strategy (Trial)	Physiological rationale	Potential drawbacks	Methodological contributions
Restrictive fluid strategy (CLASSIC, 2022)	• Positive fluid balance and fluid accumulation are associated with venous congestion, tissue edema, and organ dysfunction • Restricting further intravenous fluid after initial resuscitation might limit fluid-induced harm • Tested whether a restrictive volume strategy improves outcomes compared with standard/liberal fluid administration	• Randomization occurred after substantial fluid had already been given, so the trial tested restriction during the optimization/de-escalation phase rather than early resuscitation • The achieved separation in fluid volume between arms was modest, limiting the contrast actually tested • A population-level restrictive-vs-liberal dichotomy ignores individual fluid responsiveness, fluid tolerance, and phenotype (FR not assessed before fluid decisions) • Neutral primary outcome (no difference in 90-day mortality)	• Reinforced that a fixed restrictive-vs-liberal contrast applied uniformly may be the wrong question, shifting emphasis toward individualized fluid decisions • Large, pragmatic, multicenter design with protocolized fluid rules and explicit safety criteria • Underscored that the resuscitation phase (ROSE framework) in which a fluid strategy is applied conditions both its expected effect and the trial's validity • Provided reassuring safety data (no excess of serious adverse events with restriction)
CRT- targeted Personalized Hemodynamic Resuscitation (CRT-PHR) (ANDROMEDA-SHOCK-2, 2025)	• Sequential phenotype-driven assessment (DAP + FR + echocardiography) targeting CRT enables individualized resuscitation • CRT normalization as a dynamic stopping rule prevents unnecessary fluid escalation • Vasopressor and dobutamine tests as reversible, phenotype-targeted interventions	• CRT-PHR complexity requires trained teams and sustained implementation adherence • Echocardiography integration demands operator expertise not universally available • CRT-PHR presented lower than expected mortality in both study arms, benefit driven primarily by duration of organ support, long-term implications pending	• Provides proof of concept for a CRT-anchored, phenotype-informed bundle • Hierarchical composite outcomes (organ support duration) emerge as sensitive primary endpoints for complex resuscitation trials • Feasibility near real time data monitoring and quality assurance

ARISE, Australasian Resuscitation in Septic Evaluation; CRT, capillary refill time; PHR, Personalized hemodynamic resuscitation; CVP, central venous pressure; DAP, diastolic arterial pressure; DO₂, oxygen delivery; ED, emergency department; EGDT, Early Goal-Directed Therapy; FR, fluid responsiveness; ProCESS, Protocolized Care for Early Septic Shock; ProMISE, Protocolised Management in Sepsis; RCT, randomized controlled trial; ScvO₂, central venous oxygen saturation; VO₂, oxygen consumption

Skin perfusion in shock states represents a relevant and easily assessable microcirculatory territory, where direct clinical observation can grossly estimate the ischemia/reperfusion process as represented by the transition from cold, clammy skin to a warm, vasodilatory state with CRT normalization. The physiology behind CRT is complex and not completely understood, but it represents skin blood flow, adrenergic-induced local vasoconstriction, and microvascular reactivity [27].

A renewed enthusiasm on peripheral perfusion assessment including CRT in the early 2010's was founded in two pillars. First, relevant epidemiological and physiological observations, and an exploratory pilot study by Bakker et al. suggested a possible role of rapid peripheral perfusion normalization as a signal to safely stop fluid resuscitation [28–30]. Second, observational studies by Hernandez et al. demonstrated the rapid kinetics of CRT response to early resuscitative interventions in septic shock survivors [26]. These data were critical to challenge the traditional view of peripheral perfusion as just a marker of shock severity to a potential resuscitation target, especially in the context of lactate's limitations as described above, thus paving the way for the subsequent ANDROMEDA-SHOCK trial [5].

The ANDROMEDA-SHOCK trial compared a CRT vs. lactate-targeted resuscitation strategy for an 8 h intervention period in patients with early septic shock, testing the hypothesis that faster clinical reperfusion signaled by CRT normalization would decrease the risk of over-resuscitation, organ dysfunction, and mortality [5]. The trial demonstrated a strong trend toward decrease in 28-day mortality in the CRT group, as well as a significant impact in organ dysfunction. A posterior Bayesian analysis supported the superiority of the CRT targeted strategy [31].

Several aspects of the trial should be highlighted. First, CRT was assessed with a novel standardized technique to decrease inter-rater variability. Second, the introduction of systematic assessment of fluid responsiveness (FR) before any fluid bolus precluded administering potentially harmful fluids in 25% of patients. Third, the vasopressor and inodilator tests were applied as reversible interventions aimed at normalizing CRT and were continued only in those achieving the perfusion target as a pragmatic way to tailor hemodynamic management.

Despite the merits of the ANDROMEDA-SHOCK trial, the stepwise but somehow “one-size-fits-all” common algorithm constrained the possibility of incorporating new concepts, such as early hemodynamic phenotyping,

or finding a rational place for basic echocardiographic assessment, thus hindering progress in developing more personalized resuscitation strategies. However, the trial launched an intensive research agenda focused on CRT. Up-to-date clinical physiological studies suggest that CRT may be a key variable to assess the status of macro-to-microcirculatory coupling, the condition where improvement in systemic blood flow or MAP translates into improvements in microcirculatory flow [32]. In contrast, it appears that CRT non-response may be linked to several potential pathogenic factors, such as endothelial dysfunction and abnormal microvascular reactivity, microcirculatory dysfunction and eventually venous congestion [27, 32].

Furthermore, there at least two scenarios where CRT guided resuscitation remains unproven. First, patients who have persistent CRT derangements after structured resuscitation. Interestingly, this criterion has been proposed as one of the pillars of the SCCM/ESICM definition of refractory shock [24]. Thus, understanding how to best treat these patients requires more complex and nuanced approaches to perfusion and hemodynamic monitoring. Second, a further area of interest is why patients with normal CRT can still be vulnerable to prolonged organ dysfunction and have a considerable risk of death despite early reperfusion. Addressing these issues may contribute to a better understanding of the physiology behind this complex, yet simple, costless and universally available perfusion target.

The dilemma of the best mean arterial pressure target

Classic physiological principles position MAP as the best estimate of global perfusion pressure and identify 60 mmHg as the lower autoregulatory threshold below which organ perfusion may be jeopardized. Therefore, the EGDT trial incorporated a $\text{MAP} \geq 65$ mmHg into its resuscitation algorithm, although with limited supporting evidence [1]. Historically, this target became the de facto benchmark for both research and clinical practice. Yet, 25 years later, determining the most appropriate MAP target for individual patients remains an open question.

Under normal conditions, organ blood flow is predominantly determined by local metabolic demands [33]. Autoregulation of blood flow exhibited by most vascular beds reflects the operation of multiple mechanisms aimed to maintain a constant flow over a wide range of arterial perfusion pressures. In shock states, however, autoregulation may fail because effective input pressure to the microvascular bed becomes insufficient, not simply because systemic vasoconstriction predominates [34]. Thus, hypotension itself is a central mechanism of autoregulatory failure. Systemic hyperadrenergic vasoconstriction may further redistribute blood flow, but

its effect depends on whether increased arteriolar tone raises local resistance sufficiently to reduce the effective pressure transmitted to the microcirculation [34].

Diverse additional pathogenic mechanisms in septic shock like vasoplegia, endothelial dysfunction, myocardial depression, and abnormal microvascular reactivity further jeopardize the capacity to maintain adequate organ blood flow [33]. As a result, perfusion may decrease linearly with progressive hypotension, but with substantial variability between territories, organs, patients, and during the time-course of septic shock resuscitation. Thus, the model of one single autoregulatory threshold signaling the optimal MAP for all organs or patients represents a reductionist and overly simplistic view [35, 36].

Furthermore, MAP is a composite variable shaped by cardiac output and numerous vascular properties. Increasing MAP through vasoconstriction can, paradoxically, impair microvascular flow, since the effective tissue perfusion depends not only on MAP itself but also on its interaction with local microvascular determinants. A key determinant of these gradients is the critical closing pressure (P_{crit}), the arterial pressure below which small vessels collapse and flow ceases [37]. Functionally, P_{crit} is not a fixed anatomical pressure, but a functional, vaso-motor-dependent pressure boundary that determines the effective perfusion gradient. Thus, tissue perfusion is more precisely expressed as the tissue perfusion pressure ($\text{MAP} - P_{\text{crit}}$) rather than the conventional $\text{MAP} - \text{CVP}$ [38]. This framework helps explain why identical MAP values may be associated with different organ perfusion states across patients, organs, and phases of septic shock resuscitation. The variable and often unpredictable effects of vasopressors on these determinants may explain why raising MAP sometimes enhances, yet at other times impairs microcirculatory flow and organ perfusion but always considering the potential influence of hypovolemia on these effects.

The conventional 65 mmHg target, however, has faced repeated scrutiny. The *SEPSISPAM* trial was intended to compare higher (80–85 mmHg) versus standard (65–70 mmHg) MAP targets, hypothesizing that a higher perfusion pressure might improve outcomes [3]. Of note, MAP in the standard group was maintained at 75 mmHg well above the intended 65 mmHg target. The trial found no difference in mortality and kidney function but with a significant increase in new onset atrial fibrillation episodes in the whole population. However, chronic hypertensive patients assigned to the higher MAP target required less renal replacement therapy without this adverse effect.

As the debate evolves, it becomes clear that interpreting MAP requires a physiological perspective. Identical MAP readings can represent entirely different hemodynamic

states depending on underlying physiology. Since tissue perfusion pressure cannot be effectively measured, individualized “vasopressor tests”—brief, controlled increases in MAP—may help identify patients who truly benefit from higher MAP targets [6, 39]. Conversely, in patients with normal perfusion, it is probably safe to manage a lower MAP target as was explored by a recent study [40]. More broadly, the future likely lies in personalized, contextual, and physiology-guided titration, where changes in MAP are used as a tool to assess perfusion responsiveness rather than a target itself. However, the impact of backward pressures (CVP, intraabdominal and intracranial pressures, and so on) as determining real perfusion pressures, should always be considered.

How restrictive can we get in fluid resuscitation?

Few aspects of hemodynamic resuscitation in septic shock experienced such dramatic changes as fluid therapy during these 25 years. Indeed, the 2000's were preceded by decades of massive fluid resuscitation practice with the aim of optimizing cardiac output. EGDT followed the same trend and ended up with almost 5 L of resuscitation fluids and a CVP of 13 mmHg at the end of the 6-h intervention period [1].

However, practice shifted to a progressively more restrictive fluid resuscitation pattern during this period, driven by emerging scientific concerns and research findings. First, Malbrain et al. described the pathophysiology of secondary abdominal compartment syndrome in the 2000's [13], and the fluid overload/accumulation concept in the 2010's [41], highlighting that massive or unbalanced fluid administration could be associated with organ dysfunction, and an increased mortality risk. Second, the description of the four phases of fluid administration (Resuscitation, Optimization, Stabilization, Evacuation (ROSE) concept) provided a physiological background for more rational fluid-related interventions at the bedside but also could influence vasoactive and other decisions according to the particular phase [41–43]. More recently, a fluid stewardship strategy has been proposed to promote rational use of fluids and vasopressors and prevent iatrogenic harm through careful titration and timely de-escalation [44]. Third, the progressive penetrance of FR assessment in resuscitation algorithms allowed to focus fluid administration in effectively fluid-responsive patients. However, the FENICE study raised awareness on practice gaps related to fluid resuscitation in different critical care contexts. In fact, this global inception cohort study found that FR assessment was not routinely performed, dynamic indexes of FR were seldomly used (22%), safety limits were rarely considered, and information from previously failed fluid boluses was not always considered for further decisions [45]. The ANDROMEDA-SHOCK trial demonstrated later that

selecting the right trigger while using systematic assessment of FR avoided inadequate fluid resuscitation [46].

In this context, the CLASSIC trial addressed whether even more restrictive fluid administration in septic shock patients could further impact major outcomes [4]. The trial design faced several relevant obstacles, but most importantly, the already contemporary shift to a more restrictive practice made comparisons with usual care extremely cumbersome. Therefore, to maximize the chances of finding gradient of interventions between groups, CLASSIC applied extreme criteria to allow fluid administration in the restrictive group (e.g., MAP < 50 mmHg, lactate $\geq$ 4 mmol, and mottling score > 2). Moreover, FR was not systematically assessed during the study, nor were markers of fluid tolerance or venous congestion incorporated into treatment decisions. Thus, CLASSIC is best interpreted as testing the safety and clinical impact of an additional restrictive fluid policy in a context where usual care had already become relatively restrictive, rather than as defining optimal fluid management across diverse physiological contexts.

Finally, several contemporary reviews have highlighted that the risk of fluid-induced harm should be assessed before administering fluids independently of FR status during septic shock resuscitation. Indeed, coexisting overt or subtle venous congestions signals, vascular hyperpermeability or hypoalbuminemia among other factors, may impair fluid tolerance [47–50]. Importantly, impaired fluid tolerance and venous congestion should not be equated with cumulative fluid balance alone, since congestion may be present early despite limited fluid exposure, whereas some patients may tolerate larger volumes without evidence of congestion. Emerging trials are increasingly testing whether integration of fluid tolerance and venous congestion assessment can better define when fluids should be administered, stopped, or replaced by alternative hemodynamic interventions, and their impact on physiological and patient centered outcomes [51].

A physiology-driven iterative resuscitation algorithm: the ANDROMEDA-SHOCK-2 trial

Many drivers of progress can be identified during these 25 years (Table 2 and Fig. 3). The period was characterized, among other aspects, by the wide implementation of point-of-care ultrasound (POCUS), hemodynamic assessment by critical care echocardiography (CCE), implementation of dynamic indices of tissue perfusion, systematic evaluation of FR prior to fluid administration, and evaluation of venous congestion and fluid tolerance by several means. All these facilitated the design of more physiology-oriented resuscitation algorithms and led to expert recommendations to explore personalized resuscitative interventions in subsequent septic shock trials

Table 2 Domains of conceptual progress in septic shock hemodynamic resuscitation over the past 25 years

Domain	Key advances over the past 25 years
Tissue perfusion and the microcirculation as the core of resuscitation	• Validation of perfusion-oriented resuscitation strategies • Basic and translational research on microcirculatory dysfunction and macro-to-microcirculatory uncoupling (hemodynamic incoherence) as determinants of unsuccessful hemodynamic resuscitation • Research on the physiology, determinants, and limitations of individual perfusion-related variables • Transition of peripheral perfusion from a prognostic marker to a resuscitation target
Reappraisal of classic hemodynamic signals	• CVP: from a resuscitation target to a safety limit • MAP: from a fixed universal target to an individualized approach • Oliguria: from an exclusive marker of hypoperfusion to the potential role of venous congestion
Fluid management and stewardship	• Progressive penetrance of fluid responsiveness assessment • The ROSE concept to adapt therapeutic interventions to the physiological context • Fluid accumulation, fluid intolerance, and venous congestion as key pathogenic factors in post-resuscitative organ dysfunction • Reframing fluid resuscitation not as obligatory restriction but as better-tailored fluid administration
Vasopressor and inotropic therapy	• Better understanding of the systemic effects of norepinephrine, the rationale for its earlier use, and the emergence of alternative vasopressors • More rational and tailored use of dobutamine based on echocardiographic patterns
From fixed protocols to personalized, physiology-driven resuscitation	• Progressive and widespread use of multimodal POCUS (cardiac, venous congestion, organ flow) to guide physiology-driven hemodynamic interventions • Transition from fixed to adaptive resuscitation protocols • Recognition of basic hemodynamic phenotypes (predominant hypovolemia, vasoplegia, and cardiac dysfunction) with simple bedside tools to address physiological heterogeneity and aim at personalized resuscitation

CVP, central venous pressure; MAP, mean arterial pressure; POCUS, point-of-care ultrasound; ROSE, Resuscitation–Optimization–Stabilization–Evacuation

[52]. Accordingly, a CRT targeted, personalized hemodynamic resuscitation strategy (CRT-PHR), was designed and tested in patients with early septic shock in the large ANDROMEDA-SHOCK-2 trial [6].

CRT-PHR applied a sequential, multilayered bedside assessment, integrating CRT, pulse pressure, diastolic arterial pressure (DAP), FR tests, and early basic echocardiography to identify distinct hemodynamic phenotypes such as predominantly hypovolemic, vasoplegic or with cardiac dysfunction [53].

The trial proposed interventions according to the patient’s evolving pathophysiology and stopped protocolized intervention once CRT had normalized [6]. In

that sense, it provides one of the most comprehensive attempts to date to operationalize septic shock resuscitation as a dynamic sequence of phenotype-driven decisions.

Mechanistically, the trial is highly informative. By the end of the study period, a higher proportion of patients in the intervention arm had normalized CRT compared with usual care (86% vs. 62%). The patient trajectories show that fluids remained the major intervention, but no longer as an indiscriminate default. Sequential FR assessment followed by fluid boluses if required resulted in CRT normalization in roughly half of patients who entered the intervention group with prolonged CRT. Early CCE performed after failure of tier-1 interventions identified cardiac dysfunction in a substantial proportion of patients, while one third normalized CRT after specific treatment [6].

Another innovative aspect of CRT-PHR is the introduction of physiologically driven acute hemodynamic interventions. A low DAP triggered norepinephrine titration, explicitly targeting the vasoplegic phenotype. In patients with persistent abnormal CRT after previous steps, a MAP and/or a dobutamine test were applied, proving to be effective in a meaningful subset of patients.

ANDROMEDA-SHOCK-2 showed a statistically significant benefit for its hierarchical composite primary outcome, reported as a stratified win ratio of 1.16 (95% CI 1.02–1.33, $p=0.04$). The win-ratio compares patients between groups according to a prespecified hierarchy of outcomes, giving priority to more clinically important events (i.e. mortality) before lower-priority outcomes (i.e. hospital length of stay). In this trial, the observed benefit was mainly driven by shorter duration of vital support, whereas 28-day mortality was similar between groups and lower than expected (approximately 26.6%). Although wins were more frequent in the intervention group among patients in the high APACHE II stratum, prespecified subgroup analyses did not demonstrate formal evidence of effect modification [6].

It is important to highlight that the individual contribution of each component of the intervention algorithm remains uncertain. Indeed, because the intervention was a sequential multi-component bundle, the trial cannot isolate which element (e.g., CRT targeting, FR assessment, DAP-guided norepinephrine adjustment, the MAP/dobutamine tests, or the stopping rule) produced the effect. Therefore, the trial is best understood as the beginning of this process rather than as its endpoint, and key findings should stimulate further studies aimed at refining, validating, and expanding similarly physiology-based personalized resuscitation strategies.

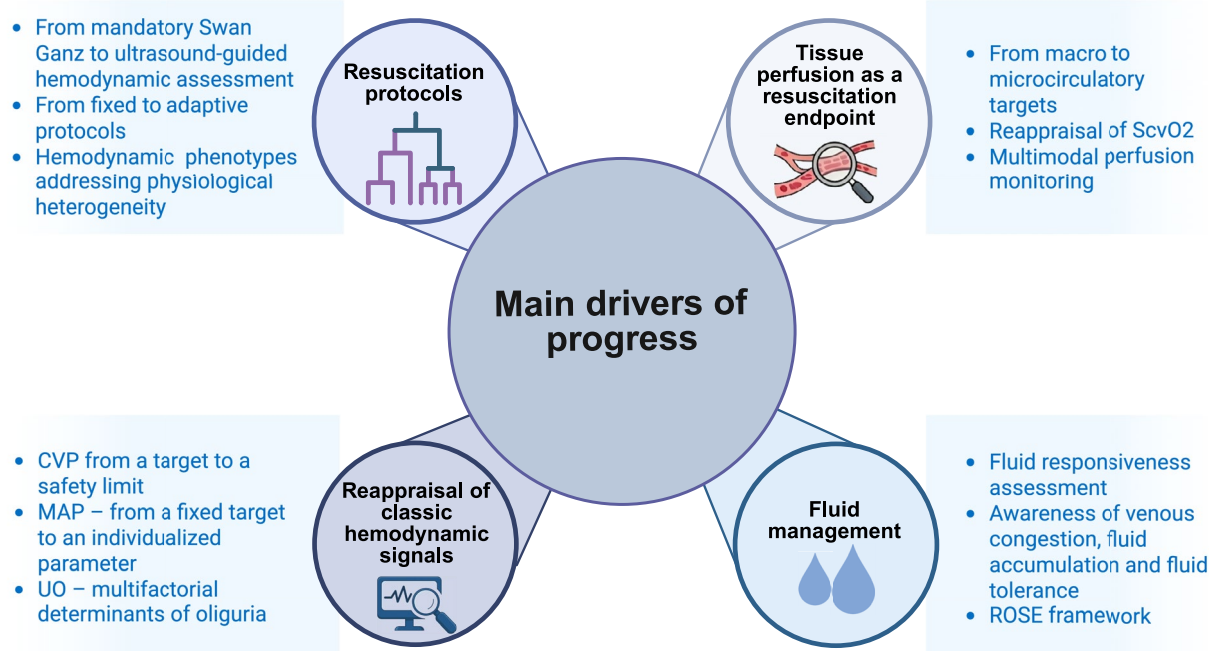


Fig. 3 Key drivers of progress in septic shock hemodynamic resuscitation over the past 25 years. The figure synthesizes four interrelated domains that have driven conceptual progress in septic shock resuscitation. First, resuscitation protocols have evolved from mandatory invasive monitoring toward ultrasound-guided hemodynamic assessment, and from fixed to adaptive, phenotype-informed protocols that address physiological heterogeneity. Second, tissue perfusion has become a central resuscitation endpoint, with a shift from macro- to microcirculatory targets, a reappraisal of ScvO₂, and the adoption of multimodal perfusion monitoring. Third, classic hemodynamic signals have been reappraised: central venous pressure has moved from a resuscitation target to a safety limit, mean arterial pressure from a fixed goal to an individualized parameter, and urine output is now interpreted through its multifactorial determinants rather than as a simple perfusion surrogate. Fourth, fluid management has been reframed around fluid responsiveness assessment, awareness of venous congestion, fluid accumulation and fluid tolerance, and the phase-based logic of the ROSE framework. Together, these drivers share a common theme: progress has been propelled less by new drugs or devices than by a deeper physiological understanding of how to apply the tools we already had. CVP, central venous pressure; MAP, mean arterial pressure; ROSE, Resuscitation–Optimization–Stabilization–Evacuation; ScvO₂, central venous oxygen saturation; UO, urine output

Reflections on 25 years of septic shock resuscitation research

A quarter-century of research, from EGD_T to the recent CRT-PHR trial, has shaped concepts and hastened progress in contemporary septic shock resuscitation. The period was characterized by a shift on the focus of resuscitation from macrohemodynamic to tissue perfusion targets, a trend that was consolidated after key studies and finally integrated into current guidelines.

In parallel, the physiological determinants of several perfusion-related variables, as well as their strengths and drawbacks were explored, leading to the conception of multimodal perfusion monitoring. More recently, broader attention has been directed to peripheral perfusion including CRT, as potential triggers and/or targets in septic shock resuscitation. This transition has been supported by consistent epidemiological data on its strong prognostic value, its universal and costless availability, its rapid kinetics of response to acute hemodynamic interventions, and the evidence generated by two recent trials.

The period was also characterized by a growing awareness on the toxicity of over-resuscitation, particularly on

fluid-related harm. The penetrance of systematic assessment of FR, the emerging concept of fluid tolerance, the use of POCUS to estimate venous congestion at the bedside, led to more rational and physiologically grounded fluid administration practices.

One unsolved problem is the profound gap between a better understanding of the pathophysiological background of the resuscitation process and the translation of this knowledge to contemporary research design and clinical practice. Some contemporary trials reflect this friction, and the contradiction of trying to address complex dilemmas with the “one size fits all” approach. Recent data suggest alternative ways to address this conundrum, essentially promoting an improved cross-talk between deep physiology and bedside monitoring and management. An example is the use of a vasopressor test, a pragmatic approach to determine if a transient and reversible increase in MAP results in a beneficial effect on tissue perfusion.

Personalization of hemodynamic resuscitation considering the extreme heterogeneity of septic shock patients is still an unfulfilled objective. However, the suggestion

that short iterative cycles of “observing, testing, intervening and reassessing” according to simple hemodynamic phenotypes may be a first step towards this aim, with the added benefit of bringing the physician closer to the bedside.

Finally, progress in septic shock resuscitation during these 25 years has been mainly driven by the translation of physiological principles to bedside care using simple and readily available clinical tools, instead of new technological or pharmacological developments. After a quarter of a century, through different resuscitation strategies, the remaining pillars are still the same, vasopressor, fluids and inotropes, yet the understanding of their limits and effectiveness has dramatically sharpened.

Conclusions

Twenty-five years of septic shock resuscitation trials reveal a consistent pattern: strategies centered on fixed and isolated physiological targets have repeatedly failed to produce reproducible improvements in patient-centered outcomes. This failure does not necessarily reflect the ineffectiveness of the interventions themselves, but rather the limitations of reductionist frameworks applied to a profoundly heterogeneous and dynamic condition. The emerging evidence supports a shift from standardized and protocolized resuscitation, toward physiology-guided, phenotype-driven form of care, in which interventions are applied as reversible tests within short decision cycles and guided by rapidly responsive perfusion signals. Trials such as ANDROMEDA-SHOCK-2 represent important steps in this ongoing transition, but further refinement, validation, and implementation across diverse settings remain necessary.

Abbreviations

RCTs	Randomized controlled trials
ScvO ₂	Central venous oxygen saturation
MAP	Mean arterial pressure
EGDT	Early goal-directed therapy
DO ₂	Oxygen delivery
ED	Emergency department
SSC	Surviving sepsis campaign
VO ₂	Oxygen demand
CRT	Capillary refill time
POCUS	Point-of-care ultrasound
CCE	Critical care echocardiography
FR	Fluid responsiveness
CRT-PHR	CRT targeted, personalized hemodynamic resuscitation strategy
ICU	Intensive care unit
CVP	Central venous pressure
SOFA	Sequential organ failure assessment
Pcrit	Critical closing pressure
SOSD	Salvage, Optimization, Stabilization, De-escalation
ROSE	Resuscitation, Optimization, Stabilization, Evacuation
DAP	Diastolic arterial pressure

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

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